

Qianqian He

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SUMMARY

University of Chicago M.S. Computer Science candidate with 3+ years of experience building AI products, agentic systems, and data platforms at Alibaba, L'Oréal, and Trina Solar. Expertise in LLMs, agentic AI, RAG, machine learning, distributed systems, software engineering and data engineering, with a track record of delivering production AI solutions and multimillion-dollar business impact.

EDUCATION

The University of Chicago

Master of Science in Computer Science

September 2025 – December 2026

Chicago, IL

Relevant Course: Machine Learning, NLP, Computer Vision, Distributed System, Parallel Computing, Advanced C++, Optimization

SKILLS

- AI/ML: AgenticAI, LLM, RAG, GenAI, Machine Learning, Deep Learning, Reinforcement Learning, NLP, CV, MLOps
- Programming: Python, C++, Java, SQL, Distributed Systems, Data Engineering, API, Database, AWS, Spark, Kubernetes, CUDA
- Frameworks & Analytics: LangChain, PyTorch, TensorFlow, Scikit-learn, Statistical Modeling, Algorithm Design, Data Analysis

PROFESSIONAL EXPERIENCE

Trina Solar Co., Ltd.

January 2024 – August 2025

Data Scientist

Shanghai, CN

- Architected and deployed an agentic supply chain optimization system leveraging LLMs, RAG, and workflow orchestration across pricing, supplier, logistics, and market intelligence data, achieving \$6.65M in annual cost savings.
- Developed forecasting, optimization, and decision-intelligence pipelines for procurement timing, supplier selection, inventory planning, and logistics optimization under dynamic market conditions.
- Built a production-scale energy management platform integrating CNN-LSTM forecasting, CPLEX optimization and reinforcement learning across 2M+ hourly time-series datasets, reducing annual operational costs by \$1.53M.
- Designed real-time data ingestion pipeline and API integration for grid, battery, and renewable energy systems, enabling adaptive dispatch optimization under demand and renewable generation uncertainty.

L'Oréal S.A.

November 2022 - December 2023

Data Analytics & Governance Manager

Shanghai, CN

- Optimized a multimodal recommendation system using transformer-based foundation models and cross-attention fusion across 10M+ structured and unstructured datasets, increasing recommendation relevance and improving user satisfaction by 13%.
- Built multimodal representation pipelines integrating purchase history, customer attributes, product metadata, images, videos, and interaction logs to improve user intent understanding and ranking performance.
- Developed a GenAI-powered analytics platform combining ETL automation, natural language querying, predictive analytics and dashboard generation, improving operational efficiency by 21.4%
- Implemented NLP-driven insight generation, conversational analytics, and automated reporting workflows to support sales performance evaluation and business decision-making.

AISHU Technology Corp.

August 2021 – October 2022

Product Manager

Shanghai, CN

- Led development of an AIOps platform using Java and Rust for anomaly detection, failure prediction, infrastructure monitoring, and root-cause analysis across distributed data center environments, improving operational efficiency by 30%.
- Designed telemetry processing pipelines and real-time monitoring services supporting servers, networks, and application infrastructure at scale.
- Implemented MLOps and backend orchestration workflows for event tracking, observability, and operational analytics, improving system scalability, reliability, and incident response.

Alibaba Group Holding Ltd.

May 2020 - July 2021

Product Manager

Hangzhou, CN

- Managed feature development and lifecycle management for Alibaba Tianchi Big Data & AI Platform adoption through user analytics, rapid prototyping, open-source datasets, and AI model deployment.
- Collaborated with engineering teams on platform architecture, testing, and CI/CD workflows to improve developer experience and platform capabilities.
- Developed a medical image classification model using ResNet34, FPN and BBN mixup in Python and Pytorch, improving macro F1 by 20.7% for benchmarking and evaluation pipelines.

PROJECTS

Multi-Agent Deep Research System | LangChain, MCP, LLMs, Reinforcement Learning

- Designed and implemented a stateful multi-agent system that automates literature retrieval, experiment generation, execution, and report synthesis for scientific discovery workflows.
- Integrated foundation models, tool calling, retrieval systems, and working-memory architectures to improve reasoning quality and agent coordination.
- Applied Active Representation Learning and Reinforcement Learning to optimize experiment selection and execution efficiency.
- Developed human-in-the-loop verification workflows and interactive interfaces to improve usability, reproducibility, and scientific research validation.

LLM-Driven Portfolio Management System | Java, Object Oriented Design, ActiveMQ, Apache Camel, LangChain

- Designed and implemented a portfolio management platform supporting portfolio construction, transaction processing, risk analysis, and performance evaluation.
- Built event-driven market data, transaction, recommendation, and portfolio management services using ActiveMQ and Apache Camel.
- Integrated LLM-powered recommendation and portfolio rebalancing workflows using LangChain and MCP based on portfolio performance, market conditions, and investor risk profiles.
- Automated investment analysis and decision-support workflows through agentic orchestration and tool-calling architectures.

High-Performance Neural Network for Image Transformation | C++, ONNX, Parallel Computing

- Designed and implemented a neural-network-based image transformation inference system in modern C++ with ONNX model generation and validation.
- Developed data ingestion, tensor encoding, pattern extraction, and transformation analysis pipelines for structured grid-based datasets.
- Built automated ONNX graph generation, model validation, and verification workflows to improve model generation and inference reliability.
- Applied parallel processing and performance optimization techniques across the end-to-end pipeline, from data loading to model verification.

Interactive Energy Harvest Simulation Application | Python, FastAPI, Docker

- Developed a distributed simulation platform for energy stabilization and harvesting using scheduling and optimization techniques.
- Built RESTful APIs, authentication services, encrypted communication, and service registration workflows using FastAPI.
- Designed hybrid state-management architecture combining in-memory storage and relational database to ensure consistency and fault tolerance.
- Implemented inter-service communication, message relaying, and automated re-registration mechanisms to support resilient distributed execution.

RESEARCH EXPERIENCE

Reinforcement Learning for Experiment Selection in Multi-Agent Scientific Discovery

Target Submission: NeurIPS 2026 Workshop

- Developed reinforcement learning methods for adaptive experiment selection in multi-agent scientific discovery systems.
- Designed consensus-based optimization and state-management mechanisms to improve experiment quality, execution efficiency, and scalability.
- Improved experiment execution success rates through monitoring, adaptive scheduling, and reinforcement learning decision policies.

Explainable AI for Reliable Energy Consumption Forecast

Accepted at EIECT 2024

- Built and evaluated machine learning forecasting models for energy consumption prediction, achieving best MSE of 0.0017 using XGBoost.
- Applied SHAP-based explainability techniques to quantify feature importance, improve model interpretability, and support model governance.

Breast Cancer Detection Using Deep Learning

NTU Business AI Lab (Distinction)

- Developed a CNN-based medical image classification model using data augmentation and regularization techniques, achieving 0.921 precision.
- Built visualization dashboards for prediction analysis, performance evaluation, and model interpretability.